
Game Theory and the Voting System

PROJECT REPORT

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Abstract

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In a collective of individuals with diverse views and thoughts, there are occasions when they need to come together to make crucial judgments between multiple options, commonly done via voting for one's favoured option. Voting theory is a mathematical framework that analyzes the features and implications of voting systems, addressing the design and analysis of the collective decision-making mechanism. Game theory, on the contrary, investigates the strategic interactions of rational agents. This project intends to explore the voting theory through the perspective of game theory, looking into the strategic conduct of voters within various voting systems. This project will discuss both single-winner and multi-winner voting systems and explain how game theory has been used, seeking to find the vulnerabilities and fortifications in various voting systems. Beyond this, this project will also suggest measures to strengthen the robustness and fairness of these systems. This endeavour seeks to enhance knowledge of one of the most essential elements of democracy.

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Chapter 1

Introduction

We all have opinions and thoughts, which, most of the time, are differing and varying from others' opinions. Furthermore, as a part of society, we come together to decide between several options. We need this to select our representatives, decide which version of a bill to enact, whether to fire a person and much more.

The most simple and causal solution for this decision problem is to take out a vote. However, there are many possible voting rules, and so this response, by itself, does not resolve the question. This gives rise to a domain of study known as voting theory or social choice theory.

Voting theory explores how groups make collective choices and the processes that assist this process. It comprises several models and conclusions that translate individual inputs like votes, preferences, and judgments into collective outputs such as choices, preferences, and welfare assessments. This field grapples with questions like how a group of people can select a winning option from a set of possibilities, the characteristics of various voting systems, the democratic nature of these systems, how collectives like electorates, legislatures, and committees can form coherent group preferences or judgments based on individual member inputs, and ways to arrange social choices in a hierarchy of social welfare.

We shall look at various single-winner and multi-winner voting rules under a mathematical framework and axioms, define a new parameter, “dissatisfaction factor”, and compare different voting systems based on this parameter.

Chapter 2

Preliminaries

2.1 The Model

There is a finite set X of potential candidates (alternatives) for a given office. The electorate is a finite set $N = \{1, \dots, n\}$ of voters. Each voter $i \in N$ is described by his utility function $u_i : X \rightarrow \mathbb{R}$. To simplify our analysis, we rule out indifference by assumption. That is, for all $x, y \in X$, if $x \neq y$, then $u_i(x) \neq u_i(y)$.

Let U_X be the set of all utility functions on X without indifference. A preference profile $u = (u_1, \dots, u_n) = (u_i, u_{-i})$ on $U(\subseteq U_X)$ is a specification of a utility function $u_i(\in U)$ for each voter $i \in N$. A ballot is a subset $Y(\subseteq X)$ consisting of candidates who are actually running for the office.

A voting rule $\mathcal{F}$ is a correspondence that, for each u and ballot Y , selects a subset $\mathcal{F}(u, Y) \in Y$ which gives the winner for all u and Y . This subset is a singleton in the case of single-winner voting rules and a non-singleton set in multi-winner voting rules.

2.2 Axioms

We now define the axioms which are standard in the voting theory field. We say that a voting rule satisfies a given axiom on domain U if the axiom holds for all the profiles u drawn on U .

2.2.1 Pareto Principle (P) on U

For all u on U , Y and $x, y \in Y$, if $u_i(x) > u_i(y)$ for all $i \in N$, then $y \notin \mathcal{F}(u, Y)$. That is, if everyone prefers x over y , then y cannot be elected.

2.2.2 Anonymity (A) on U

Fix any permutation of the electorate $\pi: N \rightarrow N$ (one-to-one and onto mapping). For any u on U , let u^π be the profile of preferences, such that, for all $i \in N$, $u_i^\pi = u_{\pi(i)}$. Then, for all Y , if $x = \mathcal{F}(u, Y)$, we have $x = \mathcal{F}(u^\pi, Y)$, given $x \in Y$.

That is, if we permute a profile, so that voter j gets i 's preferences, k gets j 's preferences and so on, the winner remains the same. In simple words, every vote is counted equally.

2.2.3 Neutrality (N) on U

Fix any ballot Y and any permutation $\rho: Y \rightarrow Y$ (one-to-one and onto mapping) of Y . For any u on U , let u^ρ be the profile, such that, for all $i \in N$, $u_i^\rho(\rho(x)) = u_i(x)$ for all $x \in Y$. Then, if $x = \mathcal{F}(u, Y)$, we have $\rho(x) = \mathcal{F}(u^\rho, Y)$.

That is, suppose we start with a profile in U and permute the candidates, the voters' utilities for the candidates are permuted correspondingly. In simple terms, all candidates are treated equally.

2.2.4 Independence of Irrelevant Alternatives (IIA) on U

For any u on U and any ballot Y , if $x = \mathcal{F}(u, Y)$ and $x \in Y' \subseteq Y$, then $x = \mathcal{F}(u, Y')$.

That is, if x is the winner from the ballot Y , then x must be the winner of the reduced ballot Y' , created by dropping some "irrelevant" candidates, given $x \in Y'$.

2.2.5 Ordinality (O) on U

For all u and u' on U , Y and $x, y \in Y$, if $u_i(x) > u_i(y) \Leftrightarrow u'_i(x) > u'_i(y)$ for all $i \in N$, then $\mathcal{F}(u, Y) = \mathcal{F}(u', Y)$. That is, the winner is determined only by the voters' ordinal ranking, and not by any cardinal information about their preferences.

2.2.6 Decisiveness (D) on U

For any Y and u , $\mathcal{F}(u, Y)$ is a single-valued function. That is, the election should result in a clear-cut winner.

Chapter 3

Voting Systems

3.1 Single-winner Voting Systems

3.1.1 Plurality (First-Past-the-Post) Rule

Mathematically, the rule is defined as: $\mathcal{F}^P(u, Y) = \{x \in Y \mid |\{i \mid u_i(x) > u_i(y) \text{ for all } x \neq y; x, y \in Y\}| \geq |\{i \mid u_i(x') > u_i(y) \text{ for all } x' \neq y; x', y \in Y\}|\}$.

In words, candidate x wins if a higher proportion of voters rank x first than they do any other candidate x' . If there are multiple such x , one is selected at random.

Consider the example election with $Y = \{A, B, C, D\}$ and $N = \{1, \dots, 30\}$ as shown in table 3.1.

For $5 + 8 = 13$ voters, candidate A is their first choice. Similarly, 9 voters chose B and another 8 voters chose C as their top choice. By plurality rule, candidate A wins. However, A is not the first choice of the majority of the voters.

3.1.2 Instant Run-off Rule

This system consist of two rounds. If a candidate get a majority, she wins. Otherwise, every other candidates except the top two candidates are ignored and the votes are recounted to find the winner.

5	8	9	8
A	A	B	C
B	C	D	B
C	B	C	D
D	D	A	A

TABLE 3.1: Example election with 4 candidates and 30 voters

Using our example, we see that A only has 13 out of 30 votes, which is less than majority. Thus, votes are recounted, this time taking only candidates A and B under consideration. Since B has 17 votes compared to 13 votes for A , B wins here.

3.1.3 Majority Rule (Condorcet's Method)

Mathematically: $\mathcal{F}^C(u, Y) = \{x \in Y \mid |\{i \mid u_i(x) > u_i(y)\}| \geq \frac{|N|}{2} \text{ for all } x \neq y; x, y \in Y\}$.

That is, candidate x is a Condorcet winner if, for any other candidate y , a majority prefers x over y .

In our example, we pair-wise compare all the four candidates to find our winner under the majority rule.

- A vs B : B beats A (13 vs 17 votes)
- A vs C : C beats A (13 vs 17 votes)
- A vs D : D beats A (13 vs 17 votes)
- B vs C : C beats B (16 vs 14 votes)
- B vs D : B beats D (30 vs 0 votes)
- C vs D : C beats D (21 vs 9 votes)

As candidate C beats every other candidate, C is the Condorcet winner of this game whereas candidate A is beaten by every other candidate, making him the Condorcet loser.

3.1.4 Rank-Order Rule (Borda Count)

Mathematically: $\mathcal{F}^B(u, Y) = \{x \in Y \mid \sum_{i=1}^N r_{u_i}(x)i \geq \sum_{i=1}^N r_{u_i}(y)i \text{ for all } y \in Y\}$, where $r_{u_i} = |\{y \in Y \mid u_i(x) > u_i(y)\}|$.

In words, candidate x 's point-score r_{u_i} for voter i with preferences u_i is the number of candidates that voter i ranks no higher than x . Candidate x is a winner if the summation of her point-scores over voters is no lower than that of any other candidate y .

In simpler terms, with $|Y|$ candidates, a candidate gets $|Y|$ points for every voter who ranks her first, $|Y| - 1$ points for a second-place vote, and so on. The candidate with the most points wins.

For the example election:

- Candidate A has $4 \times 5 + 4 \times 8 + 1 \times 9 + 1 \times 8 = 69$ points.
- Candidate B has $3 \times 5 + 2 \times 8 + 4 \times 9 + 3 \times 8 = 91$ points.
- Candidate C has $2 \times 5 + 3 \times 8 + 2 \times 9 + 4 \times 8 = 84$ points.
- Candidate D has $1 \times 5 + 1 \times 8 + 3 \times 9 + 2 \times 8 = 56$ points.

Here, candidate B wins with 91 points.

3.2 Multi-winner Voting Systems

3.2.1 Majority Approval Voting Systems

3.2.1.1 Limited Block Approval Voting

Each voter's first M block of preferences are considered to elect a committee W with size k and $M > k$. The first k out of $|Y|$ candidates with maximum votes are declared winners.

For our example and $M = 3$ and $k = 2$,

- Candidate A has $5 + 8 + 0 + 0 = 13$ votes.
- Candidate B has $5 + 8 + 9 + 8 = 30$ votes.
- Candidate C has $5 + 8 + 9 + 8 = 30$ votes.
- Candidate D has $0 + 0 + 9 + 8 = 17$ votes.

Candidates B and C are the winners.

3.2.1.2 Plurality Approval Voting

Each voter's first k preferences are considered for electing a committee W with size k . The first k out of $|Y|$ candidates with maximum votes are declared winners.

For our example and $k = 2$,

- Candidate A has $5 + 8 + 0 + 0 = 13$ votes.
- Candidate B has $5 + 0 + 9 + 8 = 22$ votes.
- Candidate C has $0 + 8 + 0 + 8 = 16$ votes.

- Candidate D has $0 + 0 + 9 + 0 = 9$ votes.

Candidates B and C are the winners.

3.2.1.3 Limited Approval Voting

Only the voter's top preference is considered for electing a committee W with size k . The first k out of $|Y|$ candidates with maximum votes are declared winners.

For our example and $k = 2$,

- Candidate A has $5 + 8 + 0 + 0 = 13$ votes.
- Candidate B has $0 + 0 + 9 + 0 = 9$ votes.
- Candidate C has $0 + 0 + 0 + 8 = 8$ votes.
- Candidate D has $0 + 0 + 0 + 0 = 0$ votes.

Candidates A and B are the winners.

3.2.2 Proportional Approval Voting (PAV) System

Proportional approval voting refers to voting methods which aim to guarantee proportional representation in case all supporters of a party approve all candidates of that party. The voters vote via approval ballots where each voter marks those candidates that the voter finds acceptable. We need to elect a committee W of size k . Let A_i denote the set of candidates considered from the voters' first k preferences for the committee.

Assuming that the voter voted for r candidates in W , the voter contributes to the score of W , the value of the r -th harmonic number, that is, $H(r) = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{r}$, where r is the number of candidates considered in the voters' first k preferences.

The PAV score of a committee $W (\subseteq Y)$ with size $|W| = k$ is defined as $SC_{PAV}(W) = \sum_{i=1}^N H(|A_i \cap W|)$. PAV selects the committee W with the maximal score.

For our example, the PAV score is calculated in table 3.2. Winners are candidates B and C .

	AB	AC	AD	BC	BD	CD
5 voters who voted AB	$5 \times (1 + \frac{1}{2})$	$5 \times (1)$	$5 \times (1)$	$5 \times (1)$	$5 \times (1)$	$5 \times (0)$
8 voters who voted AC	$8 \times (1)$	$8 \times (1 + \frac{1}{2})$	$8 \times (1)$	$8 \times (1)$	$8 \times (0)$	$8 \times (1)$
9 voters who voted BD	$9 \times (1)$	$9 \times (0)$	$9 \times (1)$	$9 \times (1)$	$9 \times (1 + \frac{1}{2})$	$9 \times (1)$
8 voters who voted CB	$8 \times (1)$	$8 \times (1)$	$8 \times (0)$	$8 \times (1 + \frac{1}{2})$	$8 \times (1)$	$8 \times (1)$
Total score	32.5	25	22	34	26.5	25

TABLE 3.2: PAV score for the example election

	P	A	N	IIA	O	D
Plurality (First-Past-the-Post)	✓	✓	✓	×	✓	✓
Instant Run-off	✓	✓	✓	×	✓	✓
Majority	✓	✓	✓	✓	✓	×
Rank-Order	✓	✓	✓	×	✓	✓
Limited Block Approval	✓	✓	✓	✓	✓	×
Plurality Approval	✓	✓	✓	✓	✓	×
Limited Approval	✓	✓	✓	✓	✓	×
Proportional Approval	✓	✓	✓	✓	✓	×

TABLE 3.3: Axioms applied to the voting systems

3.3 Axioms applied to the Voting Systems

Table 3.3 shows standard axioms defined earlier being applied to various voting systems.

Axioms P , A , and N are so natural that any sane voting systems rarely violate it. Axiom O is valid for all the systems discussed in this report due to the construct of our model, and assumptions surrounding it.

Axiom IIA highlights the phenomenon of vote splitting, which changes the outcome of the election, depending on the voting system being used.

3.3.1 Vote Splitting

Now we shall look at vote splitting, a consequence of axiom IIA . We shall consider our existing example under plurality rule to showcase this phenomenon.

Under normal circumstances, candidate A wins by 13 votes, beating B with 9 votes and C with 8 votes. Since candidate C is irrelevant to this outcome under the voting rule, we decide to drop him from the ballot.

But due to this changed ballot, B now has $9 + 8 = 17$ votes, beating A under the plurality rule. Candidate C is, therefore, significant to the original outcome and was the cause of votes being split between B and C , helping A to win.

Chapter 4

Dissatisfaction Factor

4.1 Definition

In this chapter, we introduce a new parameter dissatisfaction factor δ for a given set of Y , N , u_i on U , for all $i \in N$ and voting rule $\mathcal{F}$.

Dissatisfaction factor $\delta = \sum_{i=1}^N l_i(\mathcal{F}(u, Y))$, where l_i is the dissatisfaction index for a given u_i for every $i \in N$ and $\mathcal{F}(u, Y)$. The first k preferences are added, where k is the number of candidates to be elected.

In words, for a given preference profile u on U , ballot Y , voters N and voting rule $\mathcal{F}$, set the index of the winner 0, and going upward in the preference for i , give indices from 1 to m , where $m - 1$ is the location of winner in u_i .

4.2 Single-winner Voting Systems

We shall use the previously mentioned example and calculate dissatisfaction factor for each voting system. Here, $k = 1$. These factors are put in a bar graph in figure 4.1.

4.2.1 Plurality (First-Past-the-Post) Rule

The winner for plurality rule was candidate A . We give out indices to the preference profile as shown in the table 4.1.

Dissatisfaction factor $\delta = 0 \times 5 + 0 \times 8 + 3 \times 9 + 3 \times 8 = 51$.

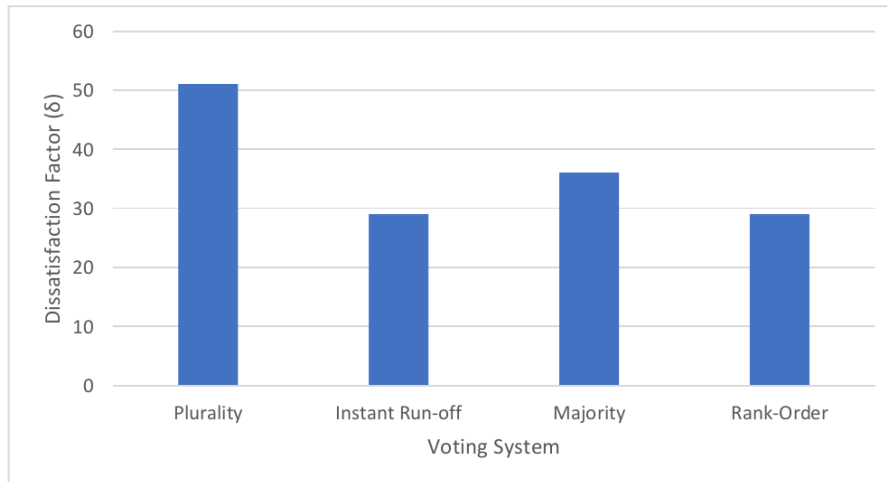


FIGURE 4.1: Dissatisfaction factor for single-winner voting systems

5	8	9	8
A (0)	A (0)	B (3)	C (3)
B	C	D (2)	B (2)
C	B	C (1)	D (1)
D	D	A (0)	A (0)

TABLE 4.1: Dissatisfaction factor for plurality rule

5	8	9	8
A (1)	A (2)	B (0)	C (1)
B (0)	C (1)	D	B (0)
C	B (0)	C	D
D	D	A	A

TABLE 4.2: Dissatisfaction factor for instant run-off rule

4.2.2 Instant Run-off Rule

The winner for instant run-off rule was candidate B . We give out indices to the preference profile as shown in the table 4.2.

Dissatisfaction factor $\delta = 1 \times 5 + 2 \times 8 + 0 \times 9 + 1 \times 8 = 29$.

4.2.3 Majority Rule

The winner for majority rule was candidate C . We give out indices to the preference profile as shown in the table 4.3.

Dissatisfaction factor $\delta = 2 \times 5 + 1 \times 8 + 2 \times 9 + 0 \times 8 = 36$.

5	8	9	8
A (2)	A (1)	B (2)	C (0)
B (1)	C (0)	D (1)	B
C (0)	B	C (0)	D
D	D	A	A

TABLE 4.3: Dissatisfaction factor for majority rule

4.2.4 Rank-Order Rule

The winner for rank-order rule was candidate B , which is same as instant run-off rule. So the dissatisfaction factor δ is 29.

4.3 Multi-winner Voting Systems

We shall use the previously mentioned example and calculate dissatisfaction factor for each voting system. Here, $k = 2$. These factors are put in a bar graph in figure 4.2.

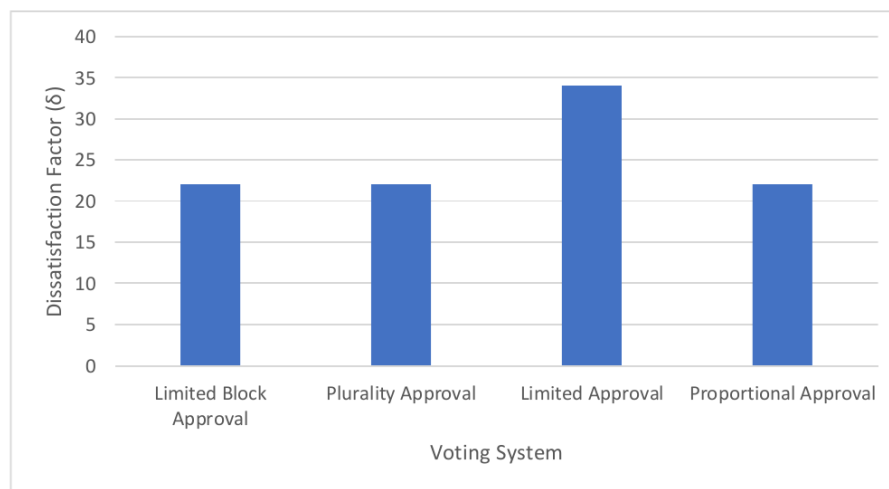


FIGURE 4.2: Dissatisfaction factor for multi-winner voting systems

4.3.1 Limited Block Approval Voting

The winner for limited block approval voting were candidates B and C . We give out indices to the preference profile as shown in the table 4.4.

Dissatisfaction factor $\delta = (1 + 0) \times 5 + (1 + 0) \times 8 + (0 + 1) \times 9 + (0 + 0) \times 8 = 22$.

5	8	9	8
A (1)	A (1)	B (0)	C (0)
B (0)	C (0)	D (1)	B (0)
C (0)	B (0)	C (0)	D
D	D	A	A

TABLE 4.4: Dissatisfaction factor for limited block approval voting rule

5	8	9	8
A (0)	A (1)	B (0)	C (1)
B (0)	C (0)	D (2)	B (0)
C	B (1)	C (1)	D (1)
D	D	A (0)	A (0)

TABLE 4.5: Dissatisfaction factor for limited approval voting rule

4.3.2 Plurality Approval Voting

The winner for plurality approval voting were candidates B and C , which is same as limited block approval voting. So the dissatisfaction factor δ is 22.

4.3.3 Limited Approval Voting

The winner for limited approval voting were candidates A and B . We give out indices to the preference profile as shown in the table 4.5.

Dissatisfaction factor $\delta = (0 + 0) \times 5 + (1 + 0) \times 8 + (0 + 2) \times 9 + (1 + 0) \times 8 = 34$.

4.3.4 Proportional Approval Voting (PAV) System

The winner for proportional approval voting system were candidates B and C , which is same as limited block approval voting. So the dissatisfaction factor δ is 22.

Chapter 5

Conclusion

In our broad examination of numerous voting systems, we have observed considerable disparities in voter satisfaction levels between single-winner and multi-winner scenarios. For single-winner voting systems, rank order (Borda count) and instant run-off voting systems indicate a decisive advantage over plurality and majority systems in decreasing voter displeasure. The rank order system, by allocating points based on voter ranks, and the immediate run-off method, by sequential elimination of the least liked candidates; both provide a more accurate reflection of voter preferences. In contrast, plurality and majority systems, like the first-past-the-post technique, frequently fail to reflect voter preferences' intricacies, resulting in more considerable unhappiness among the population.

In multi-winner voting situations, our analysis demonstrates that restricted approval voting is less successful in achieving voter satisfaction than alternative systems. While this method enables voters to approve as many candidates as seats are available, it falls short of representing the intensity and intricacies of voter preferences. Consequently, elected politicians may reflect widespread acceptability rather than the most desired alternatives, increasing voter discontent. In contrast, proportional representation systems, such as Proportional Approval Voting (PAV), effectively incorporate a more excellent gamut of voter preferences. PAV delivers a more representative and desirable conclusion by permitting voters to mark all suitable candidates and applying a score system for selection.

To conclusion, our analysis underlines the critical importance of the voting mechanism in aligning with voter satisfaction and representation goals. Rank order and immediate run-off systems in single-winner elections and proportional representation systems in multi-winner elections emerge as more successful in matching the electorate's preferences, minimizing voter unhappiness. These results underline the significance of regularly reassessing and upgrading voting methods to better reflect voters' growing needs and expectations in democratic nations.

Chapter 6

References

1. ON THE ROBUSTNESS OF MAJORITY RULE
2. Social Choice Theory (Stanford Encyclopedia of Philosophy)
3. Voting Research - Voting Theory
4. Chapter 4 Voting procedures - ScienceDirect
5. Elections and Strategic Voting: Condorcet and Borda * Partha Dasgupta and Eric Maskin
January 2020
6. [1407.3247] Computational Aspects of Multi-Winner Approval Voting
7. Tight Approximation for Proportional Approval Voting — IJCAI
8. [1602.05248] Proportional Approval Method using Squared loads, Approval removal and Coin-flip approval transformation (PAMSAC) - a new system of proportional representation using approval voting
9. Approval Voting in Practice